

Preference-Aware Coefficient Correction

for Rewarded Soups in Many-Objective NLP Alignment

Master's Thesis Proposal

Direct Preference Baseline

A simple preference-conditioned baseline sets the merge coefficients equal to the user preference:

$$\lambda = p \text{ (e.g. } p = [0.6, 0.2, 0.1, 0.1])$$

Assumes linear correspondence:

"60% weight on Objective 1 $\rightarrow$ model behaves 60% accordingly"

This Assumption Breaks Down

Objectives interact non-trivially:

- Complementary pairs: objectives reinforce each other
- Suppressive pairs: objectives interfere with each other

Note: Rewarded Soups itself searches over λ ; this thesis tests whether structure-aware correction ($\lambda \neq p$) improves on the $\lambda = p$ baseline.

Possible Examples from UltraFeedback & HelpSteer

Pair Type	Example
Complementary	Truthfulness + Instruction-Following (UltraFeedback)
Suppressive	Honesty + Helpfulness (UltraFeedback)
Suppressive	Verbosity + Correctness (HelpSteer)

Setting	Geometry	
2 objectives	Δ_2 is a line	simple sweeps are often sufficient
3-4 objectives	richer simplex geometry	complementary and suppressive relations emerge
5+ objectives	high-dimensional simplex	exhaustive search becomes resolution-dependent and uninformed

Motivation:

With more objectives, inter-objective relationships become more informative:

- **alignment:** objectives support each other
- **conflict:** objectives suppress each other

This motivates relationship-aware coefficient correction:

$$\lambda = f(p, \mathcal{R})$$

Method

1. Train N specialists from shared base model θ_0 :
 $\theta_i = \text{LocalTrain}(\theta_0, r_i)$
2. Define task vectors:
 $\delta_i = \theta_i - \theta_0$
3. Define the interpolation family:
 $\theta(\lambda) = \theta_0 + \sum_i \lambda_i \delta_i, \lambda \in \Delta$
4. Select λ by evaluating sampled or discretized coefficient vectors on validation data

Strengths

- simple and scalable
- no additional training after specialists are obtained
- supports a-posteriori preference control
- strong model-aggregation baseline

Limitations

- direct use of preference weights may ignore objective interactions
- coefficient search becomes harder in many-objective settings

Central observation:

Task vectors δ_i encode objective-specific parameter direction: $\delta_i = \theta_i - \theta_0$. Their geometry serves as a proxy for inter-objective structure (must be empirically validated).

MGDA

Finds a common Pareto-stationary descent direction by solving a min-norm problem over the convex hull of gradients.

$$\alpha^* = \operatorname{argmin}_{\{\alpha \in \Delta\}} \|\sum_i \alpha_i \nabla_{\theta} f_i\|^2$$

Property:

Inspires conflict aware coefficient balancing

PCGrad

Reduces gradient interference by projecting each task gradient onto the normal plane of any conflicting gradient.

```
if  $\hat{g}_i^T g_{\square} < 0$ :  
   $\hat{g}_i \leftarrow \hat{g}_i - (\hat{g}_i^T g_{\square} / \|g_{\square}\|^2) g_{\square}$ 
```

Property:

Removes the conflicting component with respect to the selected gradient

CAGrad

Minimizes average loss while regularizing by worst-case improvement. Generalizes GD and MGDA as special cases.

```
max_d min_i  $\langle g_i, d \rangle$   
s.t.  $\|d - g_0\| \leq c \|g_0\|$ 
```

Property:

Allows user-specified anchor g_0 ; this thesis sets g_0 from preference p .

UltraFeedback

Objective	Definition
Instruction-Following	Responds without deviating from requirements
Truthfulness	Grounded in real-world knowledge, no self-contradiction
Honesty	Expresses calibrated uncertainty, avoids fabrication
Helpfulness	Useful, correct, addresses the problem

64k prompts · 256k responses

GPT-4 based feedback

HelpSteer

Objective	Definition
Helpfulness	Overall usefulness of the response
Correctness	Factual accuracy, all relevant facts included
Coherence	Consistency and clarity of expression
Complexity	Intellectual depth and domain expertise required
Verbosity	Amount of detail — can be biased toward length

37k samples · 200 human annotators

scores from 0–4 (Likert scale)

ArmoRM — Multi-Objective Reward Modeling (EMNLP 2024)

Trains reward model with multiple interpretable objective dimensions

Output: $r(x,y) \in \mathbb{R}^k$

Uses MoE-style gating to learn context-dependent scalarization:

$$R(x,y) = g_\phi(x)^T r(x,y)$$

Relevance: ArmoRM shows multi-objective reward modeling is meaningful for NLP alignment. Its reward heads, or similar multi-objective reward models, could provide differentiable objectives r for this thesis.

DPA — Directional Preference Alignment (ACL 2024)

Models user preferences as directions in multi-objective reward space, not as scalar weights

User preference: p = preference direction in reward space

DPA motivates p as a preference direction in reward space. In this thesis, λ is the separate parameter-space merge coefficient vector.

Relevance: DPA directly motivates the distinction $p \neq \lambda$ — the central argument of this thesis.

Task Vectors:

$$\delta_i = \theta_i - \theta_0 \quad \text{and} \quad \theta(\lambda) = \theta_0 + \sum_i \lambda_i \delta_i$$

Each δ_i is a parameter-space direction induced by objective-specific fine-tuning. The Gram matrix $Q_{\text{task}} = D^T D$ is the static prior; the dynamic counterpart $Q_{\text{grad}}(\lambda)$ appears on Slide 9.

Complementary

$$\delta_i^T \delta_j > 0$$

Possible alignment signal

Possible Example:

Truthfulness + Instruction-Following

Algorithm response:

Algorithms may increase combined influence of aligned directions

Suppressive

$$\delta_i^T \delta_j < 0$$

Possible conflict signal

Possible Example:

Honesty + Helpfulness (UltraFeedback)
Verbosity + Correctness (HelpSteer)

Algorithm response:

PCGrad projects away conflict;
MGDA-style balancing may reduce the influence of conflicting directions

Central Contribution:

$$\delta_i \rightarrow \theta(\lambda) \rightarrow F(\lambda) = -r_j(\theta(\lambda)) \rightarrow \nabla_{\lambda} F(\lambda) \rightarrow \mathcal{R} \rightarrow \lambda^* = f(p, \mathcal{R}) \rightarrow \theta_{\text{merge}}$$

1. Train Specialists

Train m specialists $\theta_1, \dots, \theta_m$ from shared base θ_0 .

Compute task vectors:
 $\delta_i = \theta_i - \theta_0$

2. Initialize Soup

Set starting coefficients:

$$\lambda^{(0)} = p$$

$$\theta^{(0)} = \theta_0 + \sum_i p_i \delta_i$$

3. Compute $\nabla_{\lambda} F$

For each objective j (assuming differentiable F), compute:
 $q(\lambda) = \nabla_{\lambda} F = \delta_i^T \nabla_{\theta} f(\theta(\lambda))$
 $Q^{\text{grad}}(\lambda) = q^T q$

4. Conflict-Aware Update

Apply MGDA / PCGrad / CAGrad to $\{\nabla_{\lambda} F\} \rightarrow d^*$
 $\lambda^{t+1} = \Pi_{\Delta}(\lambda^t - \eta \cdot d^*)$

5. Final Model

$\theta^* = \theta(\lambda^*) = \theta_0 + \sum_i \lambda_i^* \delta_i$
 $\lambda^* \neq p$: corrected by inter-objective structure $\mathcal{R}$
 Evaluate vs. RS baseline

Goal: Compare different ways to compute corrected merge coefficients: $\lambda^* = f(p, \mathcal{R})$ within the same Rewarded Soups interpolation family $\theta^* = \theta_0 + \sum_i \lambda_i^* \delta_i$.

Proposed variants:

Variant	Core idea
MGDA-Soup	coefficient-space gradient balancing (secondary)
PCGrad-Soup	heuristic projection (optional ablation)
CAGrad-Soup	preference-anchored conflict-averse update (main)

Evaluation:

Primary metric: $U_p(\theta(\lambda^*)) = \sum_{p \in \mathcal{P}} p \cdot r_p(\theta(\lambda^*))$
 $U_p(\theta(\lambda_{\text{best}})) - U_p(\theta(\lambda^*))$

$\lambda_{\text{best}} = \operatorname{argmax}_{\lambda} U_p(\theta(\lambda))$

Baselines: $\lambda = p$, uniform $1/m$, Sobol/Dirichlet search, projected GD on F_p , λ_{best} (high-budget)

Note: U_p is a reward-space utility; the thesis questions whether p should equal λ in parameter space, not whether p is a valid utility.

Overarching Research Question:

Can conflict-aware and gradient-informed coefficient selection improve preference-conditioned model merging within a fixed Rewarded-Soups-style interpolation family, compared to direct preference interpolation and search-based baselines?

RQ1

How does direct preference-based coefficient selection ($\lambda = p$) behave under different inter-objective relationship structures, such as alignment, conflict, and redundancy?

RQ2

Do conflict-aware coefficient-selection methods reduce within-family regret and improve preference-weighted utility compared to direct preference interpolation and optimization-based baselines under equal budgets?

RQ3

Does conflict-aware coefficient correction preserve per-objective reward quality and reduce worst-case objective degradation compared to direct preference interpolation?

RQ4

Under which inter-objective relationship structures (alignment or suppression) does conflict-aware coefficient correction provide the largest benefit over standard Rewarded-Soups-style interpolation?

1

Literature review and thesis framing

2

Formalize the proposed method

3

Prepare data, models, and reward setup

4

Implement Rewarded-Soups baselines

5

Implement conflict-aware variants

6

Run experiments and compare methods

7

Analyze results and write thesis